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Design and Analysis of a Multisensory Line-Following Robot for Surface Cleaning and Disinfection in Indoor Environments

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Abstract: Robotics is a rapidly emerging and fascinating field in today's world. ROBOT is intelligent enough to cover the most amount of available space. Surface-disinfecting robots are a major advancement in robotics promoting cleanliness and safety in diverse environments. This paper presents the design and analysis of a multisensory line-following robot aimed at surface cleaning and disinfection. The robot integrates UV light sterilization, vacuum cleaning, AI-based object detection, and wireless charging capabilities equipped with ultrasonic, infrared, and PIR sensors. It follows a designated path while avoiding obstacles and halting UV disinfection when motion is detected for safety. The system utilizes the YOLOv3 algorithm for object recognition which alerts users to detect living beings. Experimental results demonstrate a 99% accuracy in detecting living beings, 97.68% efficiency in dust collection, and 86.67% success in obstacle avoidance, making it a promising solution for improving environmental hygiene standards.

1 Introduction

The concept and application of robotics have expanded significantly across various fields. Surface-disinfecting robots represent a significant advancement in robotics. During the COVID-19 pandemic, it became particularly hazardous for human cleaners to work in hospitals, shopping malls, houses, and other places due to the high risk of infections. Hotel workers, particularly housekeepers faced high risks of exposure [1]. A study [2] revealed higher rates of COVID-19 among environmental services workers compared with other clinical and non-clinical staff in the same institution. Autonomous surface disinfecting robots offer an effective solution to avoid human engagement in these places. Therefore, the importance of autonomous robots is increasing as people become more conscious of hygiene and cleanliness after the pandemic [3].

Surface-disinfecting robots are autonomous devices designed to sterilize or disinfect floors using technologies such as UV-C light or chemical sprays. Throughout the years, different types of robots have been developed in this field. Udoka et al. [4] reviewed Bluetooth and smart disinfectant spraying robots that helped control the spread of viruses. Ibhaize and Adekogbe[5] described an autonomous UV disinfection robot using a Raspberry Pi 4 model B, equipped with UV light for disinfection and YOLOv4 for detecting high-touch objects. A UVGI (Ultraviolet Germicidal Irradiation) robot in study [6] can operate autonomously within cramped rooms. A semi-autonomous robot called UVC-PURGE was developed [7] and evaluated for its effectiveness in inactivating the SARS-CoV-2 virus. Additionally, Korti et al. [8] focused on developing a domestic robot for waste collection, segregation, and floor cleaning.

Existing studies provide valuable insights into surface disinfection robots, but they have some limitations to consider. For example, disinfectant spraying solutions may affect the time delay

and make the system inefficient [4] which can be overcome using UV-based disinfection mechanisms [9]. The robot described in [5] may face challenges in reaching and disinfecting all surfaces in a given environment. Although some studies [6] and [7] demonstrated high accuracy in surface disinfection, precaution against UV light was not considered. To overcome these limitations, this study has presented an indoor line following mobile robot having the following objectives:

- To develop an indoor line-following mobile robot with obstacle avoidance for safe and efficient navigation.
- To integrate safe UV sterilization and vacuum cleaning mechanisms to reduce the spread of infectious microbes.
- To integrate an AI-based object detection surveillance system for enhanced monitoring and real-time alerts.
- To set up a wireless charging system to charge the robot automatically.

The robot contributes to maximizing surface disinfection through the use of UV light in a cost-effective manner. To prevent harm from UV exposure, it includes sensors that automatically turn off the UV lights when motion is detected. The robot also features AI-based object detection for efficient navigation and surveillance. Additionally, it is equipped with a vacuum cleaner and a wireless charging system.

2 Literature Review

Various works have introduced standard robots with specialized functions. Existing work by [6] and [7] explored different approaches to robotic disinfection. Paper [6] evaluated the germicidal effectiveness of a robotic UVGI device in hospital radiology settings. The robot was examined at two hospital sites with surfaces divided into eight 1×1 cm squares. The study found that UVGI reduced the microbial load by 84% on one surface and 95% on another. However, the paper did not mention any precautions to protect against UV exposure to human skin. Meanwhile, paper [7] introduced UVC-PURGE, a cost-effective semi-autonomous UVC disinfection robot developed during the COVID-19 pandemic. This robot was evaluated for its effectiveness in inactivating the SARS-CoV-2 virus and received an above-average usability score of 81.91% based on user feedback. However, it had limitations, such as reduced effectiveness in shaded or obstructed areas and the absence of UV exposure protection measures.

The RoboCoV Cleaner in paper [10] is a UV-C light disinfection robot with advanced safety features. It included dual-redundant systems such as PIR sensors and a 360-degree AI-based object detection camera to protect humans and animals. The robot needed to stay 2 to 2.5 meters from the surface and focus on each spot for 10 seconds to ensure thorough cleaning. Despite its more than 90% accuracy in safety, the robot required frequent recharging. The study in paper [11] presents a simple obstacle-avoiding robot using Arduino UNO, Bluetooth, and various

programming tools. It used an ultrasonic sensor for obstacle detection and avoided obstacles through an H-bridge-controlled movement system. The design was focused only on basic obstacle avoidance, which limits the robot's potential for other tasks like surface disinfecting or vacuum cleaning without significant modifications.

The related works in papers [8], [12], and [13] focused on robotic cleaning solutions with varying capabilities. Paper [8] introduced a domestic robot for waste collection, segregation, and floor cleaning. The robot worked in two modes: manual and automatic. In manual mode, the user controlled it via an Android app with live streaming, picking up waste with a button press, and the robot segregated it into wet or dry with 85% accuracy. In Auto mode, the robot started in a corner and cleaned in a zigzag pattern. Future enhancements using IoT or AI were suggested to make the system fully automatic. Existing paper [12] presented an intelligent household cleaning robot. The robot featured a four-wheeled drive with TT DC motors for movement and turning, an acrylic body for stability, and a detachable plastic bottle for lightweight trash collection. Advanced navigation and AI could be incorporated into future improvements. Another paper [13] described a vacuum cleaner robot with indoor mapping and positioning systems. The robot could clean for 3.7 hours with Li-ion batteries and autonomously recharge when low on battery. The robot could detect obstacles and avoid them while cleaning the area. Since it was small in size, it could accidentally be hit by unwary pedestrians. So an automated solution to this problem was planned for future versions.

We have summarized the details of the papers at a glance in Table 1:

Table 1: Brief Literature Review

Ref.	Paper Name	Contributions	Methodology	Experimental Result	Limitation
[6]	Exploring the Applicability of Robot-Assisted UV Disinfection in Radiology.	Evaluated germicidal effectiveness of a UVGI robot in hospital settings.	The germicidal effectiveness was examined at two hospital sites, with surfaces divided into eight 1×1 cm squares for sampling.	On the 2 surfaces, UVGI reduced microbial load by 84% and 95%.	No precautions were taken to protect against UV exposure while it is in proximity to human activity.

[7]	UVC-PURGE: A Novel Cost-Effective Disinfection Robot for Combating COVID-19 Pandemic.	Semi-autonomous UVC disinfection robot, focusing on cost-effectiveness and SARS-CoV-2 inactivation during covid.	Evaluated in a two-phase study: effectiveness in inactivating the SARS-CoV-2 virus and robot's usability through user feedback.	The above-average score of 81.91\% is based on user feedback.	No measures were taken to shield against UV exposure while it was near humans.
[8]	Shuchi 1.0: Robotic System For Automatic Segregation of Waste & Floor Cleaning.	A domestic cleaning robot with waste segregation and floor cleaning, operating in both manual and automatic modes.	Operated through an Android application using wireless technology with an ultrasonic sensor and Arduino board to control the movement.	Based on wet and dry segregation, accuracy was 85%.	No integration of IoT or AI technologies.
[10]	RoboCoV Cleaner: An Indoor Autonomous UV-C Disinfection Robot with Advanced Dual-Safety Systems.	A UV-C disinfection robot with dual-redundant safety systems (PIR sensors and AI-based cameras)	Utilized UV-C light for pathogen elimination and included PIR sensors and a 360-degree AI-based object detection camera.	Dual-redundant safety systems prevented UV-C light exposure, achieved over 90% accuracy.	Autonomous operation is constrained by battery life, necessitating regular recharging or battery replacement.
[11]	Design and Construction of an Obstacle Avoiding Robot Based on Arduino Platform and Programming Tools.	Created an obstacle-avoiding robot focusing on movement and obstacle avoidance	Based on electronic devices, wireless communication transmitters and sensors to operate the robot.	A robot prototype that solves the problem of obstacle avoidance.	Limits potential for additional applications like surface disinfecting or vacuum cleaning.
[12]	Autonomous vacuum cleaner.	An intelligent household	Designed with a four-wheeled TT	Ensured reliable	Lack of intelligent

		cleaning robot for trash collection.	DC motor, a detachable plastic bottle, and a coreless DC motor with high suction for cleaning.	movement with high suction from the coreless DC motor.	decision-making and adaptation power.
[13]	Designing and optimization of an autonomous vacuum floor cleaning robot.	A vacuum cleaner robot with indoor mapping, positioning, and autonomous recharging.	Utilized techniques for indoor mapping and positioning with the power of detecting and avoiding obstacles with cleaning.	Autonomous recharging capability with successful detection and avoidance of obstacles during operation.	Susceptible to being accidentally hit by pedestrians for its size.

3 Methodology

This section describes the robot's components, workflow diagram, system architecture design, circuit diagram, and pseudocodes.

3.1 Components

The components required for this project are described in Table 2:

Table 2: Component Description

Serial	Name of the Component	Description
1	4 Wheel Robot Car Chassis, BO Motors and Wheels	This includes base plate mounting holes and slots for motors and wheels. The BO motors are attached to wheels and the microcontroller Arduino or similar board to control the motor driver.
2	Arduino Uno R3	This is a microcontroller with an operating voltage of 5V. The recommended input voltage is 7-12V. There are 14 digital I/O pins and analogue input pins.
3	Motor Shield	This is designed to control DC motors and stepper motors. It is built

	L293D	around the L293D motor driver IC which allows one to control the speed and direction of two DC motors independently.
4	HC-SR501 PIR Motion Sensor	Passive Infrared (PIR) motion sensor that detects movement by sensing the infrared radiation emitted by objects.
5	Servo Motor	It is a rotary actuator or linear actuator that allows for precise control of angular or linear position, velocity, and acceleration.
6	Ultrasonic Sensor HC-SR04	This module has two eye-like projects in the front which form the Ultrasonic transmitter and receiver. It uses sonar to determine the distance.
7	UV LED	This is a type of LED that emits light in the ultraviolet spectrum. They have various applications due to their ability to produce UV light efficiently and reliably.
8	Short Shaft 180 DC Motor	This is used with an operating voltage range of 3V-9V which can be operated using two A4 batteries or with a 5V USB power supply.
9	IR Sensor	It has a pair of infrared transmitting and receiving tubes. When the transmitted light waves are reflected, the reflected IR waves will be received by the receiver tube.
10	5V Relay Module	Relay module serves as an interface for transferring information between systems. It connects the vacuum cleaner to the battery. When the robot follows the line and returns to home position, the vacuum automatically turns off.
11	Wireless Charging Module	The Wireless Charging Module is applied to automatically charge the robot after returning to the home.
12	20A Battery Management System	A 3S 2P lithium-ion battery pack is used. A 20A BMS protects the battery pack from overcharging, discharging, and overheating.
13	MT3608 DC-DC Step-Up Converter Module	The BMS (Battery Management System) is connected to the wireless charging module. The charging module has an output of 5V and 2A, but as the BMS circuit requires an input of 12.6V, it is needed to increase the voltage using this module.
14	LM2596 STEP DOWN DC-DC Buck Converter	The vacuum cleaner motor operates on 9V, but the BMS outputs 12V, so a converter is used to reduce the voltage.
15	ESP-32 Cam Module	It is used for object and living being detection, as well as for the surveillance system.

16	QC 3.0 2.0 USB charging module	To charge the ESP32-CAM.
17	Bluetooth Module	It is used to send the sensor values to the mobile app, showing if the robot has returned to the initial point and if any motion has been detected.
18	Vacuum Cleaner's Fan	The fan in a vacuum cleaner generates suction power allowing the device to collect dust and debris from surfaces.

3.2 Workflow Diagram

The overall flowchart of the proposed system is shown in Figure 1. It starts by determining whether the robot is at its home location. If the robot is not at home, it proceeds with its functionalities. Once the tasks are completed, the robot returns home for charging.

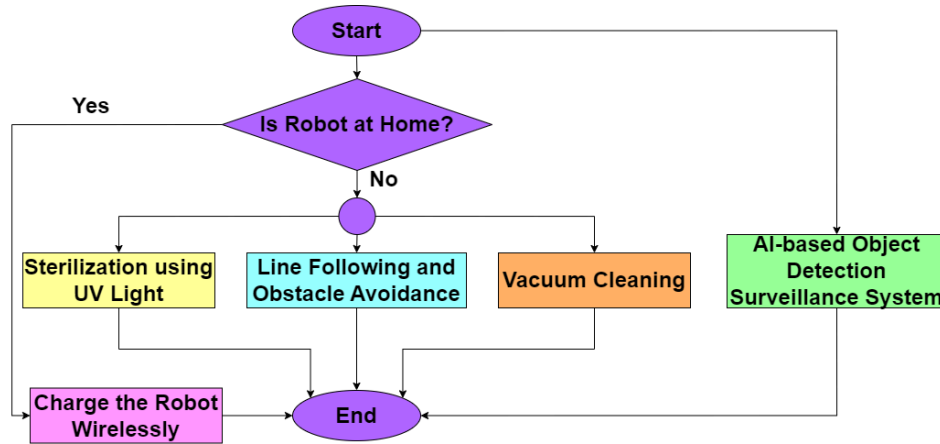


Fig 1: Workflow Diagram of the Presented System

The whole method is divided into four major parts:

- Line-Following and Obstacle Avoidance:** The robot initiates the line-following process using ultrasonic and IR sensors and checks if both IR sensors detect a black surface, indicating it is at home. If not, it follows a designated path by constantly monitoring ahead. If an obstacle is detected within 20 cm, the robot immediately halts. To avoid the obstacle, it checks the distances on both the left and right sides. If right distance is greater than or equal to the left distance, the robot turns right, moves forward, and then turns left to continue on its path. Conversely, if the left distance is greater, it turns left, moves forward, and turns right to resume its path. At the end of its task, if the robot arrives at home, it stops and sends a status update to the app via Bluetooth. The workflow diagram of this part is shown in Figure 2.

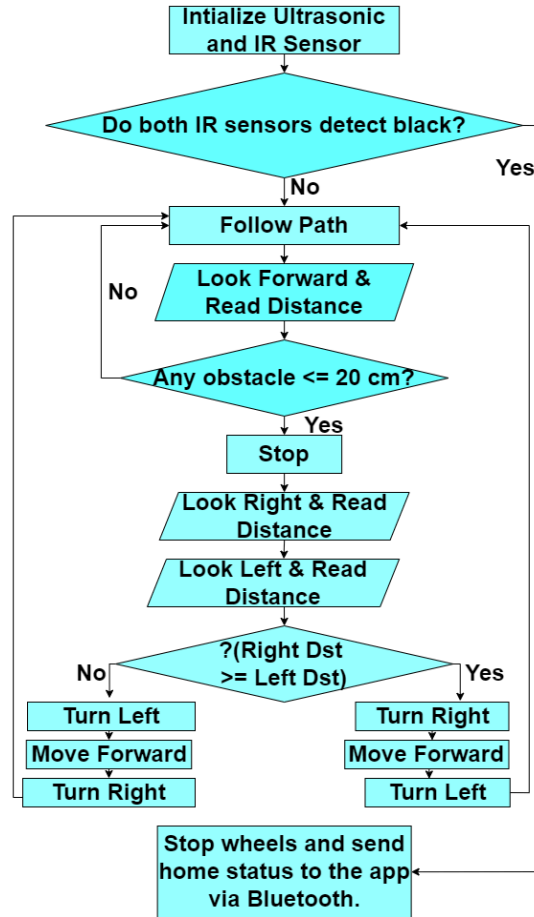


Fig 2: Line-following and Obstacle Avoidance

- b. **Sterilization Using UV Light:** UV lights are activated when no motion is detected. A PIR motion sensor continuously monitors the surroundings. However, if motion is detected, the system turns off the UV light for safety. After turning off the UV light, it sends the motion detection status to a connected application through Bluetooth. Finally, the system implements a 3-second delay before restarting the process. The workflow diagram of the sterilization part is shown in Figure 3.

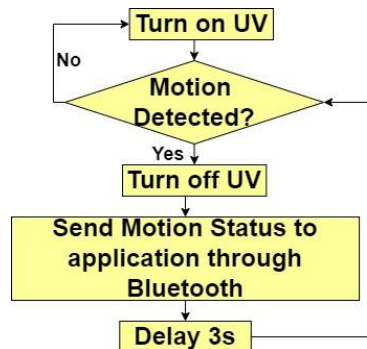


Fig 3: Sterilization Using UV Light

- c. **AI-based Object Detection and Surveillance System:** The robot utilizes the YOLOv3 object detection algorithm to identify and track objects in real-time. The system continuously scans the environment for objects. If no object is detected, it continues scanning. When an object is detected, it is shown on a display for the user to view. If the detected object is a living being, a warning email is sent to the user. After sending the alert, the system pauses for 60 seconds before resuming the scanning process. The workflow diagram of this part is shown in Figure 4.

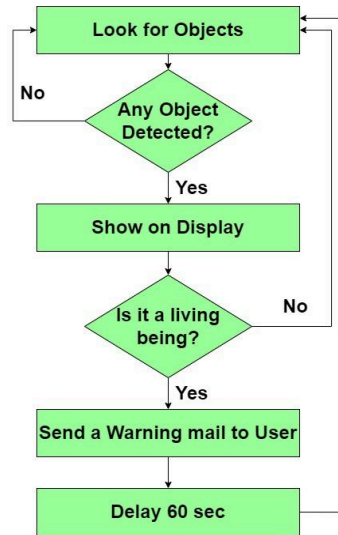


Fig 4: AI-based Object Detection and Surveillance System

- d. **Vacuum Cleaning:** In vacuum cleaning mode, the robot activates its fan motor to start the vacuuming process. The fan creates suction to pick up dust and debris from the floor. This vacuuming function operates in parallel with other tasks to ensure efficient cleaning of the designated area. If the robot returns to its home, the vacuum cleaner automatically turns off until it is needed again. The workflow diagram of the vacuuming part is shown in Figure 5.

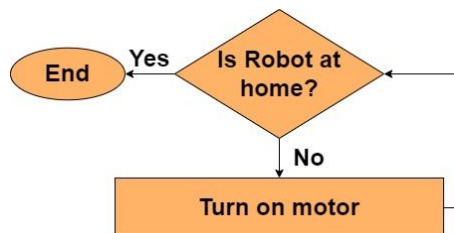


Fig 5: Vacuum Cleaning

3.3 System Architecture

The robot employs multiple sensors and components including IR sensors, PIR sensor, ultrasonic sensor, servo motor, UV lights and vacuum cleaner. Data from the sensors is processed by a central microcontroller which communicates via Bluetooth with a mobile phone. The phone receives data and warnings about detected objects from ESP32 CAM module and forwards this information to the user. If the object is a living being then a warning is sent via email over the internet. Additionally, a wireless charging module powers the system's sensors and motors. This design integrates smart cleaning, motion detection, UV disinfection, and real-time communication with the user. The overall system architecture is presented in Figure 6.

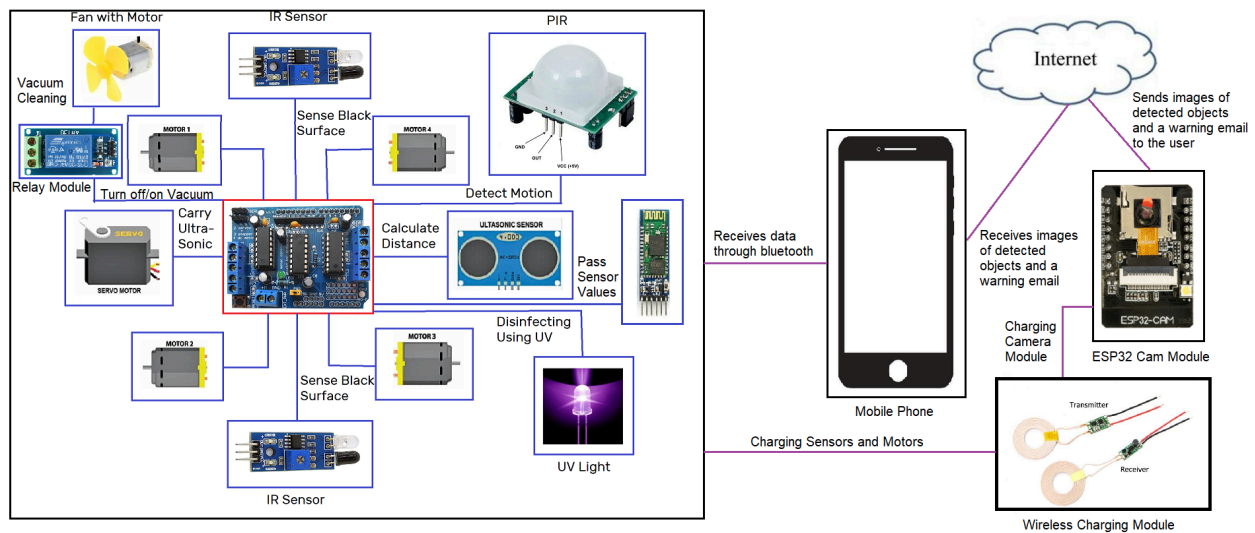


Fig 6: System Architecture

3.4 Circuit Diagram

The circuit diagram consists of an operational unit, a sterilization and processing unit, and a charging unit. In the operational unit, IR sensors, four BO motors, one ultrasonic sensor, and a servo motor have been connected to an Arduino UNO. In the sterilization and processing unit, PIR sensors, UV lights, and a Bluetooth module have been connected to another Arduino UNO. The microcontroller in the operational unit has been grounded with the microcontroller in the processing unit. Both the operational and processing units have been powered by the charging unit. The charging unit includes six lithium-ion batteries configured in a 3S2P arrangement (three in parallel, two in series) and these batteries have been connected to a BMS circuit. The circuit diagram is given in Figure 7.

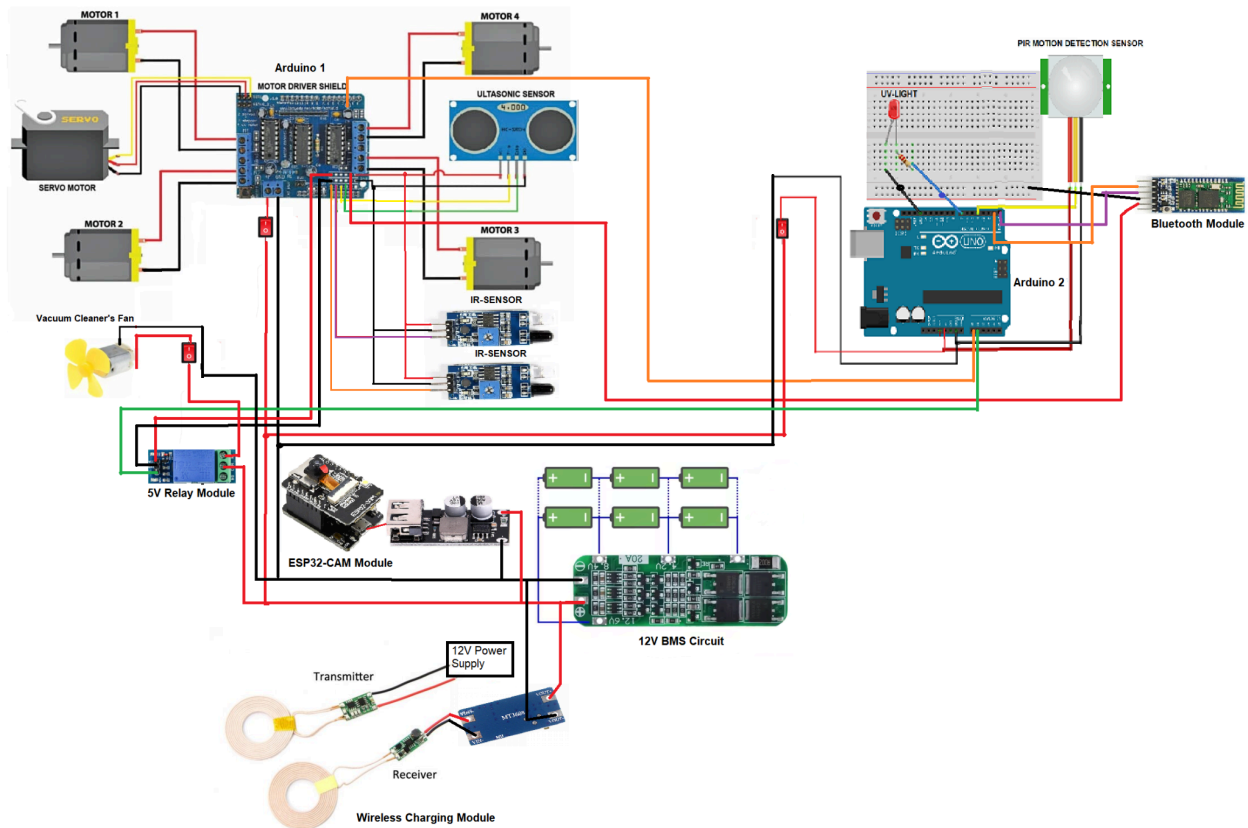


Fig 7: Circuit Diagram

3.5 Algorithm

a) Line Following with Obstacle Avoidance

```

function line_following_with_obstacle_avoidance():
(1)  initialize variables and constants
(2)  setup robot
(3)  while (true) do
(4)    if (Both IR sense Black) then
(5)      objectAvoid()
(6)    else if (Left IR senses black AND Right IR senses white) then
(7)      objectAvoid()
(8)      moveLeft()
(9)    else if (Left IR senses white AND Right IR senses black) then
(10)     objectAvoid()
(11)     moveRight()
(12)    else if (Both IR sense white) then
(13)      stop()
(14)    end if

```

```

function objectAvoid():
(16)  distance ← getDistance()
(17)  if (distance ≤ 20) then
(18)    stop()
(19)    lookLeft()
(20)    lookRight()
(21)    if (rightDistance ≤ leftDistance) then
(22)      moveLeft()
(23)      moveForward()
(24)      moveRight()
(25)      if (Right IR senses black) then
(26)        line_following_with_obstacle_avoidance()
(27)      else
(28)        moveForward()
(29)    else
(30)      moveRight()
(31)      moveForward()
(32)      moveLeft()
(33)      if (Left IR senses black) then
(34)        line_following_with_obstacle_avoidance()
(35)      else
(36)        moveForward()
(37)    end if
(38)  else
(39)    moveForward()
(40)  end if

```

b) Surface Sterilization Using UV Light and Motion Sensing

```

function PIRmotionSensor():
(1)  pirState ← LOW
(2)  val ← 0

(3)  Set ledPin as OUTPUT
(4)  Set motionPin as INPUT

(5)  while (true) do
(6)    val ← Read(motionPin)
(7)    if (val = HIGH) then
(8)      Write(ledPin, LOW)
(9)      if (pirState = LOW) then
(10)        Print("Motion detected!")
(11)        pirState ← HIGH
(12)      end if
(13)    else
(14)      Write(ledPin, HIGH)
(15)      if (pirState = HIGH) then
(16)        Print("Motion ended!")
(17)        pirState ← LOW
(18)      end if
(19)    end if
(20)  end while

```

c) AI-based Object Detection and Surveillance System:

```
function detect_living_being():
(1) url ← 'http://192.168.39.138/cam-hi.jpg'
(2) modelConfig, modelWeights ← load YOLOv3 configuration and weights
(3) net ← initialize_network(modelConfig, modelWeights)
(4) while (True) do
(5)   img ← capture_image(url)
(6)   blob ← create_blob_from_image(img)
(7)   net.setInput(blob)
(8)   outputs ← net.forward(outputNames)
(9)   living_being ← detect_objects(outputs, img)
(10)  if (living_being) then
(11)    image_path ← save_detected_image(living_being, img)
(12)    send_email(living_being, image_path)
(13)  end if
(14)  display_image(img)
(15)  if (exit_condition) then
(16)    break
(17)  end if
(18) end while
end function

function send_email(living_being, image_path):
(1) current_time ← time.now()
(2) if (current_time - last_email_time < 60) then
(3)   return
(4) end if
(5) email ← create_email(subject, body, image_path)
(6) try:
(7)   smtp_server.connect()
(8)   smtp_server.send(email)
(9)   last_email_time ← current_time
(10)  print "Email sent for living_being"
(11) catch:
(12)  print "Failed to send email"
end function
```

4 Environmental Setup

The robot follows a black line path which is needed to be 3.5 cm thick for accurate tracking. The prototype of the robot is shown in Figure 8 and the path followed by the robot is presented in Figure 9. The gap between the paths needs to be 12 inches or 30.48 cm.

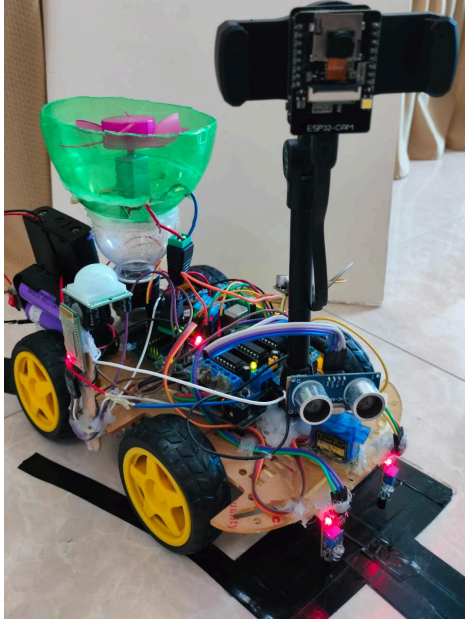


Figure 8: Robot Prototype

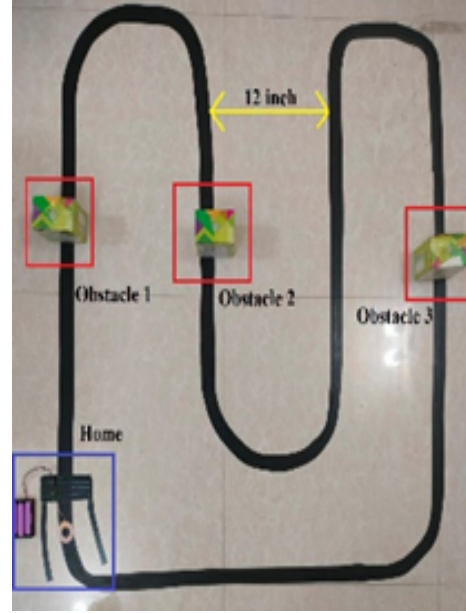
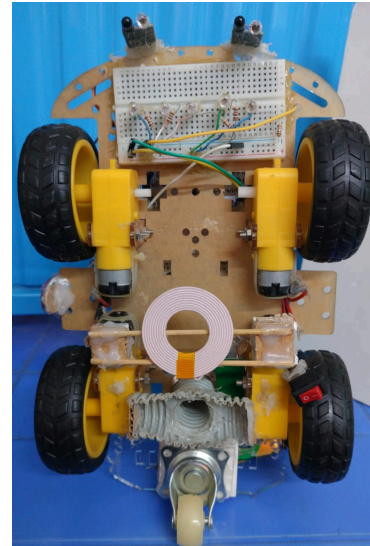


Figure 9: Example Scenario of path with the robot

When both IR sensors detect black line at the same time, the robot is back in the home position. It stops moving, turns off the UV lights and vacuum cleaner and starts charging using a wireless module. The charging station has a transmitter coil and a receiver coil underneath it shown in Figure 10.



(a)



(b)

Fig 10: (a) Charging station at home (Transmitter coil), (b) Receiver coil of charging module attached to the robot

A simple vacuum cleaner built into the robot is shown in Figure 11. The robot also uses UV light for sterilization, which turns on when the PIR sensor detects no movement. It has five UV lights connected in parallel, each with a wavelength of 100-400 nm and a brightness of 2500 MCD. The UV light setup is shown in Figure 12. A mobile application called ‘SanitiBot’ has also been developed using Flutter to show the robot’s real-time status and camera view. The robot’s status and camera view in the app are shown in Figure 13.

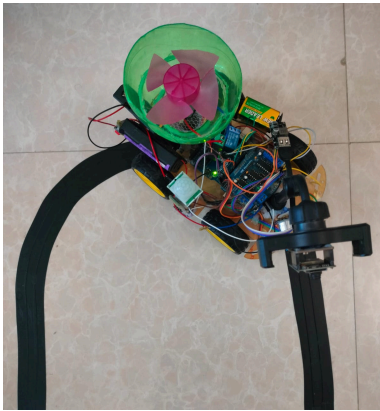


Fig 11: Vacuum Cleaning

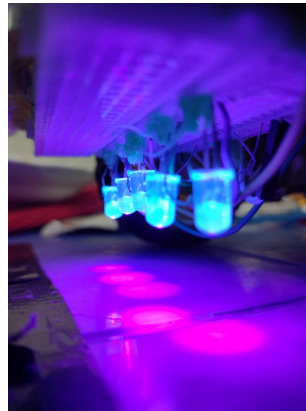


Fig 12: UV Light Set-Up

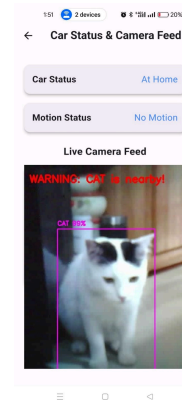


Fig 13: Mobile App Interface

The Arduino Integrated Development Environment (IDE) is an essential tool used in this project for programming and controlling the robot. The Arduino IDE provides a user-friendly interface that allows for easy coding, debugging, and uploading of sketches (programs) to the Arduino microcontroller.

The robot has a 4-wheel motor system and needs 7.4 volts to operate at optimal speed. It uses UV light for sterilization which turns on when the PIR sensor detects no motion. If motion is detected, the UV light turns off and can be reactivated after 3 seconds. The robot has five UV lights connected in parallel, each with a wavelength of 100-400 nm and a brightness of 2500 MCD. To turn on the UV light, two 3.7-volt batteries are needed, providing a total of 7.4 volts. In vacuum cleaning mode, the robot activates a 9-volt fan motor to create suction, picking up dust and debris as it moves. This function operates alongside the other tasks.

The primary purpose of this project is surface sterilization and vacuum cleaning. To maximize the coverage area, a line-following feature is implemented. The robot cannot roam freely and must always follow a designated path.

5 Experimental Result

The robot equipped with YOLOv3 has been tested in our environment and showed satisfactory results. Each detection of a living being successfully triggered an email alert, demonstrating the robot’s reliability in real-time monitoring and alerting. The detection process is displayed in

Figure 14 and the email alert is displayed in Figure 15. It achieved 99% accuracy in detecting living beings and 98% accuracy for other objects, as shown in Figure 16.



Figure 14: Living Being Detection

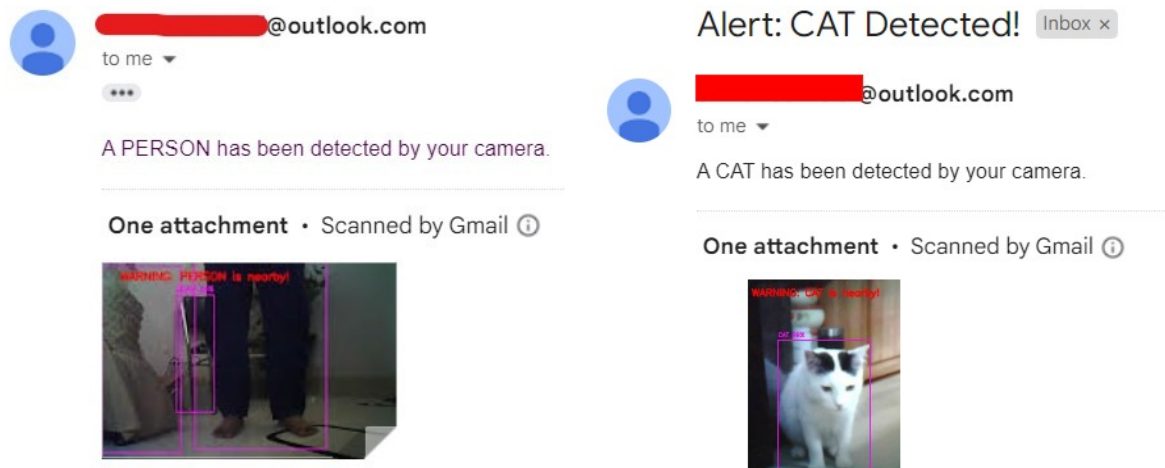


Figure 15: Warning email received after living being detection

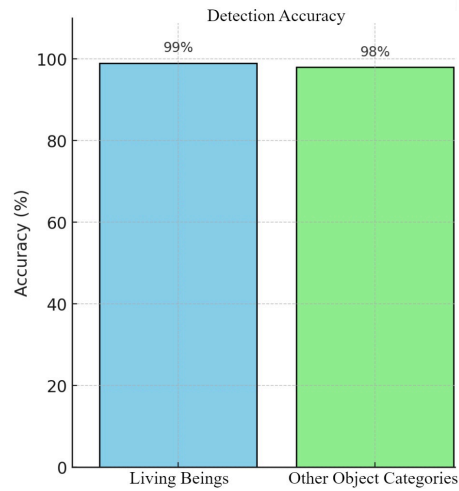


Fig 16: Accuracy of detecting living beings and objects

The graph in Figure 17 shows how the PIR motion sensor interacts with the UV light control system with a 3-second delay. When motion is detected (a), the PIR sensor turns off the UV light for 3 seconds shown by the drops in the LED state (b). During this time, the sensor ignores any further motion, as seen by the gaps between the spikes of motion detection (a). This ensures that people or animals are kept safe from harmful UV exposure.

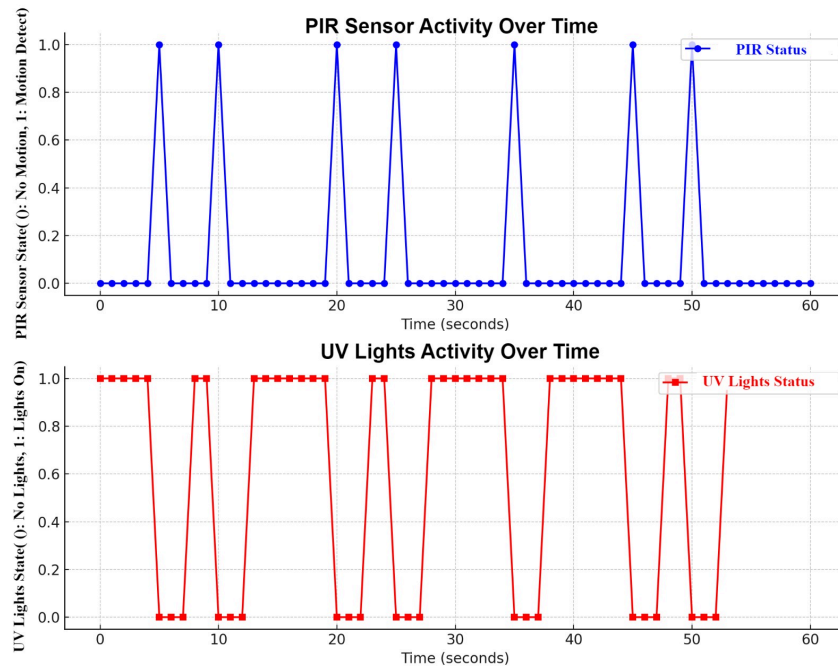


Fig 17: Correlation Between PIR Motion Detection and UV Light Activation With 3-Second Delay Over Time

The robot's vacuum cleaning performance has been evaluated by comparing the initial and collected dust weights across five trials. The dust collection percentage has been calculated by dividing the collected dust weight by the initial dust weight for each trial and then averaging the results. The overall data collection percentage was 97.68 %. The results are presented in Table 3.

Table 3: Experimental Result of Vacuum Cleaning

Initial Weight (g)	Collected Weight (g)	Dust collection percentage
0.0157	0.0152	96.82%
0.0151	0.0141	93.38%
0.0164	0.0164	100.00%
0.0157	0.0156	99.36%
0.0172	0.0170	98.84%
	Average	97.68%

It is also tested to see whether the robot can avoid obstacles successfully or not. In our designed environment, we have three obstacles. The robot is tested five times and the success rate is 86.67%. The experimental results are shown in Table 4.

Table 4: Experimental Result of Obstacle Avoidance

Total number of Obstacles	Number of Avoided Obstacles	Obstacle Avoidance Percentage
3	2	66.67%
3	3	100%

3	2	66.67%
3	3	100%
3	3	100%
	Average	86.67%

The total area of the sample environment is approximately 3,104.89 cm², calculated using a path width of 3.5 cm, a gap of 30.48 cm between each path, and 3 loops. The total width for all 3 loops is 101.94 cm (33.98 cm per loop, including the gap), and the height of the path is 30.48 cm. With the robot's width being 16 cm, it will cover the entire area as it moves across the paths and gaps efficiently.

6 Impact

In this section, we have discussed the environmental, economic, and social impacts of the robot.

6.1 Environmental Impact

Chemical disinfectants dominate the cleaning and disinfection sector particularly in healthcare and general sanitation [14]. These chemical disinfectants can be harmful for both the environment and human health. The robot reduces the need for chemical disinfectants by integrating UV sterilization and vacuum cleaning functionalities. The UV light setup is located on the front lower side of the robot. First, the dust is sterilized using UV light, and then the vacuum cleaner picks it up. This process is beneficial to the environment because sterilizing dust before suction helps eliminate harmful microorganisms and reduces the spread of contaminants. The robot also ensures efficient energy consumption by activating the cleaning process when necessary.

6.2 Economical Impact

The development of this robot aims to be cost-effective so that it can be accessible to a wide range of users, including hospitals, offices, and public spaces. The initial investment in the robot can lead to significant long-term savings by reducing labor costs associated with manual cleaning. The UV light and vacuum cleaning mechanisms reduce the need for frequent replacements of cleaning disinfectants, thereby lowering operational expenses. Additionally, the robot can prevent the spread of diseases, which will reduce healthcare costs due to infections acquired at work. The affordability and efficiency of the robot make it a viable option for

businesses and institutions looking to improve their cleaning protocols without going over budget.

6.3 Social Impact

The robot reduces human exposure to harmful cleaning agents and environments by performing automatic cleaning. It ensures safety and prevents accidental UV exposure as PIR sensors deactivate UV lights when they detect human presence. The robot's ability to maintain high cleanliness standards can contribute to overall public health by reducing the transmission of infectious diseases. This is particularly beneficial in high-traffic areas such as hospitals, schools, and offices. This improvement in public health can lead to increased productivity and well-being for positive social change.

7 Limitation and Future Work

Several enhancements can be implemented to improve the system's overall performance and functionality. Currently, the robot follows the lines to cover the maximum area, but this approach limits adaptability to different environments as lines may not always be present. To address this, a gyroscope sensor can be used to enable accurate angular turns, and a wheel encoder can help ensure precise distance coverage by counting wheel rotations. Additionally, integrating a LiDAR sensor in future can create a 2D map of the space and follow an efficient cleaning path using a smart algorithm. Adopting a round-shaped robot design would significantly increase surface coverage. Adding roller and side brushes to the vacuum cleaner would improve cleaning efficiency in corners and edges. Reducing camera latency for real-time object detection is another critical area for improvement. Integrating AI-driven algorithms could further optimize object detection and decision-making processes.

8 Conclusion

The development of the multisensory line-following robot provides an efficient and autonomous solution for surface cleaning and disinfection. By integrating UV sterilization and vacuum cleaning, the robot ensures effective microbial reduction while maintaining user safety through motion detection systems. The use of AI for object detection improves its operational capabilities and makes it suitable for various environments such as hospitals, offices and public spaces. Experimental results showed high performance in both cleaning and object detection, with improvements in energy efficiency due to its wireless charging system.

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